

Implementation notes for the Julia policy game

What changed from the first example

The initial example was written in Python because the local data and plotting libraries were already available. That was expedient, but inconsistent with the language of the surrounding project and less useful for extending the original Julia code. The substantive implementation is now entirely in Julia.

The other correction is economic rather than cosmetic. Previously, the fiscal-dominance chart changed the weights in the authorities' objective functions. That treated the regime as if it were a change in tastes. In the corrected model, preferences are fixed primitives. Fiscal dominance is instead represented by active/passive policy rules.

Fixed primitives and changing solution concepts

The central bank's loss is

$$L_M = \sum_{t=0}^{T-1} \beta^t [\pi_{t+1}^2 + 0.25x_{t+1}^2 + 0.12(u_t - u_{t-1})^2 + 10^{-4}u_t^2].$$

The fiscal authority's loss is

$$L_F = \sum_{t=0}^{T-1} \beta^t [0.25\pi_{t+1}^2 + x_{t+1}^2 + 0.45b_{t+1}^2 + 0.08(g_t - g_{t-1})^2 + 10^{-4}g_t^2],$$

with `beta = 0.99`. These expressions are used unchanged in every case.

- **Nash:** each authority chooses its entire 20-quarter instrument path taking the other path as given.
- **Cooperation:** a planner minimises `0.5 L_M + 0.5 L_F`. The 50–50 number is a planner aggregation weight, not a change in either primitive objective.
- **Central-bank leadership:** the central bank commits to a path after substituting the fiscal authority's full best-response function.
- **Fiscal leadership:** the fiscal authority performs the symmetric exercise.

Because transitions are linear and losses quadratic, the paths are found by solving linear first-order conditions. No numerical optimiser is required.

Policy-rule regimes

The fiscal-dominance exercise is intentionally separate from the strategic solution concepts. The same macro parameters and same two losses evaluate both regimes.

Rule parameter	Monetary dominance	Fiscal dominance
Nominal rate response to inflation	1.50	0.80
Fiscal tightening response to debt	0.35	0.00
Monetary debt accommodation	0.00	0.10

The first column represents active monetary/passive fiscal policy: the nominal rate satisfies the Taylor principle and fiscal policy stabilises debt. The second represents passive monetary/active fiscal policy: the nominal inflation coefficient is below one and fiscal policy does not adjust to debt.

Since the model's monetary instrument is the **real** rate gap, its inflation coefficient is the nominal coefficient minus one. Consequently, passive monetary policy can produce a declining real rate when inflation rises. The additional debt-accommodation coefficient makes the servicing-cost channel explicit.

Current Australian estimates

Parameter	Estimate	Meaning
rho_x	0.629	Output-gap persistence
sigma_i	0.0369	Real-rate effect on the output gap
chi_g	0.0514	Public-demand effect on the output gap
rho_pi	0.563	Inflation-gap persistence
kappa	0.155	Phillips-curve output coefficient
Demand residual SD	0.464	Output-gap percentage points
Supply residual SD	0.670	Annualised inflation percentage points

The IS regression has 107 usable quarterly observations and the Phillips curve has 106. Ordinary least squares currently produces the required economic signs; the Julia code checks those signs and stops if a future data vintage reverses one. It does not silently clip coefficients.

Reading the output

For the one-standard-deviation demand contraction, the fixed-preference Nash joint loss is approximately 0.179, compared with 0.168 under cooperation. This is the free-riding mechanism in the original draft: both instruments can support demand, so each authority has an incentive to leave more adjustment to the other.

For an inflation shock, the fiscal authority initially supports activity while the central bank tightens, so the two instruments partly offset. Leadership changes which authority carries more of the adjustment, but the

loss ranking is conditional on the calibration and shock.

In the policy-rule fiscal stress experiment, monetary dominance reduces the debt gap from 1.0 initially to about 0.445 after 20 quarters. Under fiscal dominance it remains around 0.668, because fiscal policy does not respond to debt and monetary policy is passive. Both paths are evaluated with the same loss functions.

Scope

The implementation nests the strategic accommodation channel of fiscal dominance, not the full fiscal theory of the price level. The detailed boundary between the three-equation core, the strategic game and the FTPL extension is set out in [THREEEQUATIONMODEL.md](#).

Generated from MODEL_NOTES.md. Model and outputs are reproduced by `julia --project=. run_model.jl`.